

THASANKA KANDAGE

tkandage.com | 587-598-6558 | thasanka.kandage@gmail.com | linkedin.com/in/thasanka-kandage | github.com/ThasankaK | Canadian Citizen

EXPERIENCE

Software Developer Intern

Jan 2026 – Dec 2026

Ericsson

Ottawa, ON

- Built and deployed a LangGraph/FastAPI AI agent with a TypeScript and Tailwind chat UI; 1,000+ conversations
- Integrated Jira, Gerrit, and Jenkins APIs for ticket processing, build/delivery requests, and CI health checks
- Cut developers CI context switching from 5-10 minutes to seconds by automating routine workflows
- Helped developers troubleshoot and find guidance via chat backed by a 3M+-record PostgreSQL vector store
- Designed a two-stage intent router using System One models to dispatch requests to specialized sub-agents
- Built hybrid RAG over 3M+ records using HNSW, keyword search, and RRF; cut retrieval from ~20s to <1s
- Deployed CronJobs to fetch/embed ~100 Jira tickets/week into PostgreSQL and summarize CI loop failures
- Wrote Dockerfiles and Kubernetes manifests to deploy the frontend, backend, and CronJobs

AI Engineer Intern

Sep 2025 – Oct 2025

Customer Maps

Calgary, AB

- Built an Express.js backend on Google Cloud Run for a Shopify AI extension for e-commerce clients
- Integrated agents built on Gemini Enterprise Agent Platform and the Shopify MCP server into the extension
- Improved responses with prompt engineering and in-agent state management to track user preferences

Research Assistant

May 2025 – Aug 2025

University of Alberta

Edmonton, AB

- Created an autonomous drone survey routing system using 3D A* and nearest-neighbor optimization
- Cut flight time by 46% while keeping 91% coverage vs. default DJI routes; co-authored ISARC 2026 paper

Research Assistant

Sep 2024 – Aug 2025

University of Alberta

Edmonton, AB

- Ran multi-objective optimization of solar thermal salt storage tanks using genetic algorithms in MATLAB
- Modeled trade-offs of 5 design parameters while optimizing for energy capture efficiency and thermal uniformity
- Published and presented the work as a first author on a conference paper at ASME ES 2025

Machine Learning Engineer Intern

Jan 2024 – Aug 2024

Hub for Neuroengineering Solutions

Lethbridge, AB

- Trained CNN and Vision Transformer models, reaching 93% mAP@50-95 on mice imagery
- Reduced manual image labeling time by building an active-learning tool that retrains on user corrections
- Clustered image embeddings with t-SNE and DBSCAN to surface gaps in the training dataset
- Selected high-impact outliers via convex-hull analysis, improving dataset diversity

PROJECTS

Software Team Member — UAlberta RoboMaster | C++, OpenCV, Python

Sep 2024 – Aug 2025

- Wrote real-time C++/OpenCV armor-plate detection for the team's auto-aiming system
- Built Python camera calibration tooling; placed 9th/20 in 3v3 at RoboMaster NA 2025

UFC Fight Dataset | Python, BeautifulSoup, Pandas

- Scraped and published an open UFC fight dataset; 1,500+ views, 300+ downloads

EDUCATION

University of Alberta

Expected Apr 2027

B.Sc. Computer Engineering, Software Co-op

Edmonton, AB

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, TypeScript/JavaScript, Bash, MATLAB

AI/ML: PyTorch, TensorFlow, scikit-learn, LangGraph, Langfuse

Frameworks: FastAPI, Express.js, OpenCV, Pandas, NumPy, BeautifulSoup, ROS2

Infrastructure: Docker, Kubernetes, Rancher, Helm, Kaniko, GitLab CI/CD, Jenkins, Argo CD, JFrog, Git, Linux

Cloud & Data: AWS S3, Google Agent Platform, Google ADK, Cloud Run, PostgreSQL, MongoDB, Kafka